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$$\lim_{n \rightarrow +\infty} \sqrt{n+1} - \sqrt{n} = +\infty - (+\infty) \Rightarrow \text{onbepaald}$$

$$\lim_{n \rightarrow +\infty} \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}}$$

$$= \lim_{n \rightarrow +\infty} \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}}$$

$$= \lim_{n \rightarrow +\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{+\infty} = 0$$

$$\text{adh } x_n = \{0\}$$

$$\liminf_{n \rightarrow +\infty} x_n = \limsup_{n \rightarrow +\infty} x_n = 0$$

References